# DOMiNANCE CVS MOCK EXAMINATION 03

**Cardiovascular Physiology | 60 questions | 90 minutes | 60 marks**

Choose the single best answer (A-D) for each question. The answer key is on the page immediately after the last question.

**1. Which sequence correctly traces normal electrical conduction from atria into ventricular myocardium?**

A. AV node -> SA node -> left ventricle -> right atrium.
B. SA node -> AV node -> His bundle -> Purkinje fibres.
C. SA node -> Purkinje fibres -> AV node -> His bundle.
D. Purkinje fibres -> atria -> SA node -> AV node.

**2. A very light stroke briefly leaves a pale line on the skin. How is this reaction best classified?**

A. A brief local vasoconstrictor response.
B. The red line of local capillary dilation.
C. The spread of an axon-reflex flare.
D. The wheal component of the triple response.

**3. A patient with raised intracranial pressure develops hypertension and bradycardia together. What best explains the slow pulse?**

A. The hypertension directly activates cardiac beta-1 receptors.
B. The atrial volume reflex reacts to the sympathetic pressure rise.
C. The arterial baroreflex reacts to the sympathetic pressure rise.
D. The cardiac pacemaker stops responding to vagal signals.

**4. A sudden increase in sympathetic venous tone can support cardiac output mainly through which immediate effect?**

A. Venoconstriction shifts stored blood centrally.
B. It dilates veins, increasing blood stored in the limbs.
C. It creates new circulating plasma within seconds.
D. It closes arteriolar valves and stops peripheral outflow.

**5. Why can vision fail before consciousness during strong positive acceleration?**

A. Retinal perfusion falls as blood shifts toward the feet.
B. The retina receives blood at higher pressure than the heart.
C. Intracranial pressure exceeds local arterial pressure.
D. Positive acceleration shifts blood toward the head.

**6. What most clearly differentiates myocardial infarction from brief stable angina?**

A. Stable angina involves transient irreversible injury to the perfused myocardium.
B. Infarction pain is usually relieved promptly by stopping exertion.
C. Infarction involves brief ischaemia without permanent myocardial cell damage.
D. Prolonged ischaemia can produce myocyte necrosis.

**7. During ventricular diastole, what would an increase in peripheral resistance do to the rate of aortic pressure decline if compliance were unchanged?**

A. Reverse arterial flow from capillaries toward the aorta.
B. Leave the decline unaffected by downstream resistance.
C. Slow the decline by reducing downstream runoff.
D. Speed the decline by increasing downstream runoff.

**8. Immediately after rapid loss of a large volume of whole blood, haemoglobin concentration is unexpectedly unchanged. Why?**

A. Plasma enters faster than red cells are lost.
B. Red cells and plasma were lost together.
C. Erythropoietin instantly makes new red cells.
D. Red-cell mass increases enough to compensate.

**9. A drug primarily dilates large systemic veins rather than arterioles. Which immediate circulatory change is most likely?**

A. Peripheral resistance rises sharply in the arterioles.
B. More blood is stored in the venous reservoir.
C. Capillary smooth-muscle contraction becomes stronger.
D. More blood is forced towards the right atrium.

**10. Why can xanthines such as theophylline produce a positive inotropic effect at the signalling level described in the lecture?**

A. They directly prevent L-type calcium-channel opening.
B. They accelerate the breakdown of intracellular cAMP.
C. They reduce the breakdown of intracellular cAMP.
D. They block cardiac beta-1 receptor activation upstream of cAMP formation.

**11. After cord clamping and lung aeration, what pressure change favours functional closure of the foramen ovale?**

A. Pulmonary resistance remains very high.
B. Umbilical venous inflow increases right atrial pressure.
C. Left atrial pressure exceeds right atrial pressure.
D. Right atrial pressure rises above left atrial pressure.

**12. During a stress response, circulating adrenaline dilates vessels in working skeletal muscle while constricting many skin vessels. Why?**

A. Skin tissue receives a much lower concentration of adrenaline.
B. Different vascular beds contain markedly different amounts of plasma.
C. The muscle vessels lack any adrenergic receptors.
D. Different vascular beds express different adrenergic receptors.

**13. A low-frequency third heart sound occurs in early diastole. What ventricular event coincides with it?**

A. Opening of the aortic valve.
B. Rapid passive ventricular filling.
C. Isovolumetric ventricular contraction.
D. Atrial depolarisation before systole.

**14. Which response becomes particularly important for supporting pressure during sustained rather than momentary upright posture?**

A. Renin-angiotensin effects and renal salt retention.
B. Suppression of aldosterone-driven sodium retention.
C. Loss of sympathetic arteriolar responsiveness.
D. Persistent inhibition of all vascular reflexes.

**15. A patient has hydrocephalus due to impaired absorption by arachnoid villi without an obstructed ventricular pathway. What type is this?**

A. Communicating hydrocephalus.
B. Noncommunicating hydrocephalus.
C. Isolated ventricular haemorrhage.
D. Obstructive aqueductal hydrocephalus.

**16. What helps maintain forward systemic arterial flow between heartbeats?**

A. Capillaries generate contractions that propel blood upstream.
B. The diaphragm pumps blood through the aorta during inspiration.
C. Venous valves directly generate aortic pressure.
D. The elastic recoil of large arteries sustains diastolic flow.

**17. A recording shows that cardiac vagal stimulation lowers the firing rate of the sinoatrial node. Which additional effect is expected?**

A. Stronger ventricular contraction through beta-1 activation.
B. Faster pacemaker discharge through alpha-1 activation.
C. Faster impulse conduction through the AV node.
D. Slower impulse conduction through the AV node.

**18. A ventricular depolarisation vector runs almost perpendicular to the viewing axis of a lead. What should that lead record?**

A. A prolonged ventricular depolarisation interval.
B. A small or biphasic ventricular deflection.
C. A tall upright ventricular deflection.
D. A tall downward ventricular deflection.

**19. A patient with severe tricuspid regurgitation has a prominent systolic jugular venous wave. Which wave becomes especially large?**

A. x descent.
B. v wave.
C. a wave.
D. y descent.

**20. In severe haemorrhagic shock, why does sympathetic constriction of splanchnic vessels help initially?**

A. It redistributes flow and mobilises stored venous blood.
B. It increases hepatic oxygen extraction to 100%.
C. It increases flow to the intestinal mucosa above baseline.
D. It prevents all intestinal ischaemia during shock.

**21. How can impaired coronary perfusion create a vicious cycle during severe shock?**

A. Myocardial hypoxia lowers arterial resistance enough to compensate.
B. Low coronary blood flow increases myocardial ATP stores.
C. Reduced stroke volume directly increases coronary perfusion pressure.
D. Coronary ischaemia weakens contraction, further reducing output.

**22. A patient's right ventricular pumping fails and right atrial pressure rises. Which consequence is most likely?**

A. A larger pressure gradient for systemic venous return.
B. Reduced congestion in peripheral systemic veins.
C. Reduced venous return gradient with congestion.
D. Complete disappearance of the jugular venous pulse.

**23. A small additional deflection appears after the T wave in an otherwise interpretable trace. Which description fits the U wave?**

A. Atrial depolarisation occurring before the next P wave.
B. A possible late ventricular repolarisation phenomenon.
C. Rapid ventricular depolarisation immediately before QRS.
D. A ventricular plateau interval between QRS and T.

**24. Why does positioning an accelerating pilot more horizontally improve tolerance to positive g?**

A. It increases the distance between the heart and brain.
B. It shortens the heart-to-brain hydrostatic column.
C. It raises the capillary filtration rate in the legs.
D. It directs pooled blood toward the ankles.

**25. A ventricular impulse occurs independently of atrial excitation after complete AV conduction block. What has happened?**

A. A subsidiary escape pacemaker has taken over.
B. Atrial excitation now travels directly through the lungs.
C. The aortic valves transmit an alternate impulse.
D. The SA node has increased its firing rate.

**26. Which substance passes relatively readily across the intact blood-brain barrier without a carrier?**

A. Plasma albumin.
B. Carbon dioxide.
C. A protein-bound drug.
D. A large immunoglobulin.

**27. During exercise, sympathetic activity increases throughout the body, yet working muscle receives much more blood. Which explanation is best?**

A. Local metabolic dilation outweighs constriction in active muscle.
B. Systemic arteriolar constriction raises every tissue's flow equally.
C. A higher systemic pressure redirects flow equally among tissues.
D. Local metabolic constriction outweighs dilation in active muscle.

**28. A patient loses substantial albumin into urine during kidney disease. What Starling-force change is expected?**

A. Increased red-cell viscosity causing filtration.
B. Increased plasma colloid osmotic pressure.
C. Reduced plasma oncotic pressure promotes oedema.
D. Reduced capillary permeability everywhere.

**29. A patient becomes dizzy on standing after starting a sympatholytic drug. Which compensatory response is most directly weakened?**

A. Constriction of systemic arterioles and veins.
B. Contraction of calf muscles during walking.
C. Dilatation of cerebral vessels by carbon dioxide.
D. Flow of blood through competent venous valves.

**30. Cardiac beta-adrenergic stimulation raises coronary flow overall mainly through which chain?**

A. Direct beta stimulation -> closure of coronary arterioles.
B. More contraction -> less ATP use -> less adenosine.
C. More work -> metabolites -> arteriolar dilation.
D. Less contraction -> less oxygen use -> greater vascular tone.

**31. In a whole-body Fick measurement, which blood should the return-side oxygen sample represent?**

A. Blood in a single peripheral venule.
B. Capillary blood from active skeletal muscle only.
C. Mixed venous blood entering the lungs.
D. Oxygenated blood in a systemic artery.

**32. A patient has had high arterial pressure for months. Why might baroreceptors fail to drive pressure back to the original baseline?**

A. Their stretch sensitivity increases indefinitely without adaptation.
B. Their afferent pathways turn into motor pathways over time.
C. Their signals stop entering the medulla during hypertension.
D. Their operating range resets around the sustained level.

**33. Why is a distinct atrial repolarisation wave usually absent from a normal ECG?**

A. Atrial repolarisation coincides mainly with the ST segment, where it is too small to detect.
B. Atrial repolarisation generates no electrical voltage differences.
C. Atrial repolarisation occurs just before the SA node starts the next beat.
D. Its small electrical signal is obscured by the ventricular QRS complex.

**34. Which condition is most likely to depress myocardial contractility rather than enhance it?**

A. Raised circulating catecholamines.
B. Severe hypoxia with acidosis.
C. Increased calcium availability in cells.
D. Acute beta-1 adrenergic activation.

**35. A stiff ventricle has high filling pressure but a preserved ejection fraction. What is the primary defect?**

A. Increased contractility and ejection.
B. Impaired relaxation and filling.
C. Reduced electrical activation of the SA node.
D. Excessive ventricular compliance.

**36. A patient with coronary atheroma is comfortable at rest but develops exertional chest pain. Which explanation fits best?**

A. The narrowed artery fully dilates to match any increase in cardiac demand.
B. A faster heart rate prolongs the diastolic time available for coronary flow.
C. Sympathetic stimulation lowers myocardial oxygen need during exercise.
D. Restricted coronary flow reserve cannot meet increased demand.

**37. What does the normal jugular venous 'v' wave mainly represent?**

A. Atrial filling while tricuspid is closed.
B. Closure of the pulmonary semilunar valve.
C. Right atrial electrical depolarisation.
D. Completion of left ventricular ejection.

**38. Electrical stimulation of the medullary pressor area increases arterial pressure. What is the principal neural pathway?**

A. Direct stimulation of the carotid sinus sensory fibres.
B. Direct inhibition of firing in the sinoatrial pacemaker.
C. Excitation of sympathetic neurons in the spinal cord.
D. Inhibition of sympathetic neurons in the spinal cord.

**39. A standard ECG uses ten skin electrodes but produces twelve electrical views. How is this possible?**

A. The limb electrodes are relocated to the chest during every cardiac cycle.
B. Several leads are calculated from combinations of the same electrode voltages.
C. Two extra electrodes inside the heart are automatically activated for recording.
D. Each electrode produces one lead, with two more produced by blood-pressure sensors.

**40. Why do Purkinje fibres help the two ventricles contract as coordinated pumps?**

A. They create the calcium plateau in ventricular working myocytes.
B. They slow ventricular excitation until the semilunar valves can open.
C. They generate most of the delay between atrial and ventricular activation.
D. They rapidly activate ventricular myocardium.

**41. During standing, head-level arterial pressure falls by 30 mmHg and cerebral venous pressure falls by 25 mmHg. How does arterial minus venous driving pressure change?**

A. It falls by about 25 mmHg.
B. It rises by about 5 mmHg.
C. It rises by about 55 mmHg.
D. It falls by about 5 mmHg.

**42. An irregular tracing contains eleven QRS complexes in a six-second span. What is the estimated average ventricular rate?**

A. 66 beats/min.
B. 110 beats/min.
C. 90 beats/min.
D. 100 beats/min.

**43. Beta-1 adrenergic stimulation generally increases ventricular contractility through which sequence?**

A. Reduced cAMP -> less calcium entry -> weaker contraction.
B. Increased vagal tone -> reduced pacemaker slope -> stronger contraction.
C. Increased cAMP/PKA signalling -> greater calcium availability.
D. Blocked ryanodine receptors -> larger systolic calcium release.

**44. Why can some bronchial venous drainage slightly lower oxygen content of blood reaching the left atrium?**

A. Bronchial venous drainage enters only systemic veins, avoiding pulmonary mixing.
B. Bronchial arteries arise from the pulmonary trunk and deliver deoxygenated blood.
C. All bronchial venous drainage rejoins systemic arteries before entering the lungs.
D. Some bronchial venous blood joins pulmonary veins.

**45. What can follow when the right ventricle fails to deliver adequate forward flow to the lungs?**

A. Pulmonary flow rises despite low right ventricular output.
B. Less blood reaches the left heart for systemic output.
C. Systemic tissues receive more oxygen-rich blood immediately.
D. Left ventricular filling necessarily increases sharply.

**46. What event ends isovolumetric contraction and begins ventricular ejection?**

A. Ventricular pressure exceeds the outflow artery pressure.
B. Atrial contraction briefly increases end-diastolic volume.
C. Ventricular pressure falls below aortic pressure.
D. Ventricular pressure falls below atrial pressure.

**47. A student calls the entire PR interval "the AV nodal delay." What important part of the interval does this description miss?**

A. The delay between ventricular systole and valve opening.
B. The period during which the ventricular myocardium is repolarising.
C. The time between the end of QRS and the start of T.
D. The spread through atria and conduction towards the ventricles.

**48. A brief sharp pain raises a patient's pressure, but prolonged severe pain later causes faintness. What best accounts for the difference?**

A. Early venous pooling gives way to strong vasoconstriction.
B. A brief pressor phase gives way to vasodilation.
C. Long pain increases total peripheral resistance without limit.
D. Brief pain reduces sympathetic output from central pathways.

**49. Why may the aortic component of the second heart sound become more prominent in systemic hypertension?**

A. Brisk aortic valve closure under higher pressure.
B. Earlier tricuspid closure following atrial contraction.
C. Late mitral valve closure during ventricular relaxation.
D. Reduced pulmonary resistance delaying right ventricular ejection.

**50. A patient with left-sided heart failure feels breathless when lying flat but improves after sitting up. Why?**

A. Sitting up increases venous return from the legs.
B. Sitting up directly increases pulmonary capillary pressure.
C. Lying flat shifts peripheral blood toward congested lungs.
D. Lying flat reduces central blood volume.

**51. Arterial pressure increases suddenly and baroreceptor firing rises. What changes first in central reflex control?**

A. The SA node starts generating impulses without modulation.
B. Sensory traffic to the medulla decreases despite added stretch.
C. NTS activation increases pressor and spinal sympathetic firing.
D. NTS activation increases depressor and vagal signalling.

**52. How can fluid enter a blind-ended lymphatic capillary when interstitial pressure rises?**

A. A central lymphatic heart pumps blood into tissues.
B. A continuous layer of smooth muscle forces plasma outward.
C. Overlapping endothelial flaps admit interstitial fluid.
D. Rigid endothelial junctions prevent all protein entry.

**53. Where does the ductus arteriosus carry much fetal right ventricular output?**

A. From umbilical vein into inferior vena cava.
B. Directly from right atrium to left atrium.
C. From left ventricle into hepatic portal vein.
D. From pulmonary trunk into descending aorta.

**54. A person becomes angry during an argument, and both pulse and pressure rise. Which route can initiate this response?**

A. Cortical signals acting through the hypothalamus on autonomic centres.
B. Atrial filling signals acting through the kidneys on increased urine output.
C. Cerebral CSF signals acting through the ventricles on the heart.
D. Platelet serotonin acting through capillaries on cardiac pacing.

**55. A local inflammatory mediator increases postcapillary venular permeability. What occurs?**

A. More protein and water can escape, promoting tissue swelling.
B. Capillary filtration stops until arterial pressure rises.
C. The plasma protein concentration increases within the vessel.
D. The vascular wall becomes impermeable to water.

**56. What role do arteriovenous anastomoses serve in fingers and other exposed skin?**

A. They prevent sympathetic regulation of skin perfusion.
B. They transfer blood directly into lymphatic capillaries.
C. They provide adjustable shunts for heat exchange.
D. They divert blood away from skin when core body temperature rises.

**57. A measured QRS occupies 3.5 small squares at 25 mm/s. Which statement is supported by the lecture's usual duration range?**

A. It lasts 0.10 s and is at the upper edge of the stated range.
B. It lasts 0.07 s and is within the stated usual range.
C. It lasts 0.14 s and is longer than the stated usual range.
D. It lasts 0.35 s and is longer than the stated usual range.

**58. Where is the largest controllable fall in systemic vascular pressure normally produced?**

A. Within the right atrium alone.
B. Across large elastic arteries.
C. Across the venae cavae.
D. Across resistance arterioles.

**59. Which combination correctly characterises isovolumetric ventricular contraction?**

A. All valves closed, ventricular pressure rises, volume fixed.
B. All valves closed, ventricular pressure falls, volume fixed.
C. AV valves open, semilunar valves shut, volume falls.
D. AV valves open, semilunar valves open, volume rises.

**60. What closes the aortic valve at the beginning of ventricular relaxation?**

A. Venous pressure exceeds aortic pressure.
B. Atrial pressure rises above left ventricular pressure.
C. Aortic pressure exceeds ventricular pressure.
D. Left ventricular pressure rises above aortic pressure.

# ANSWER KEY

Mock 03 - 60 Questions | One mark each

Check your answers only after completing the examination.

| Q01-Q15 | | Q16-Q30 | | Q31-Q45 | | Q46-Q60 | |
|---|---|---|---|---|---|---|---|
| 01. | B | 16. | D | 31. | C | 46. | A |
| 02. | A | 17. | D | 32. | D | 47. | D |
| 03. | C | 18. | B | 33. | D | 48. | B |
| 04. | A | 19. | B | 34. | B | 49. | A |
| 05. | A | 20. | A | 35. | B | 50. | C |
| 06. | D | 21. | D | 36. | D | 51. | D |
| 07. | C | 22. | C | 37. | A | 52. | C |
| 08. | B | 23. | B | 38. | C | 53. | D |
| 09. | B | 24. | B | 39. | B | 54. | A |
| 10. | C | 25. | A | 40. | D | 55. | A |
| 11. | C | 26. | B | 41. | D | 56. | C |
| 12. | D | 27. | A | 42. | B | 57. | C |
| 13. | B | 28. | C | 43. | C | 58. | D |
| 14. | A | 29. | A | 44. | D | 59. | A |
| 15. | A | 30. | C | 45. | B | 60. | C |

Answer key ends here. Review explanations separately in the question-bank archive.